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Plancherel–Polya-type inequalities for entire functions of exponential type in $L_p(\mathbb{R}^d)$

Isaac Pesenson

Department of Mathematics, Temple University, Philadelphia, PA 19122, USA

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Abstract

The goal of the paper is to prove generalizations of the classical Plancherel–Polya inequalities in which point-wise sampling of functions (δ-distributions) is replaced by more general compactly supported distributions on $\mathbb{R}^d$. As an application it is shown that a function $f \in L_p(\mathbb{R}^d)$, $1 \leqslant p \leqslant \infty$, which is an entire function of exponential type is uniquely determined by a set of numbers $\{\Psi_j(f)\}$, $j \in \mathbb{N}$, where $\{\Psi_j\}$, $j \in \mathbb{N}$, is a countable sequence of compactly supported distributions. In the case $p = 2$ a reconstruction method of a Paley–Wiener function f from a sequence of samples $\{\Psi_j(f)\}$, $j \in \mathbb{N}$, is given. This method is a generalization of the classical result of Duffin–Schaeffer about exponential frames on intervals.
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1. Introduction

A classical theorem of Plancherel–Polya states that if a function $f \in L_p(\mathbb{R})$, $p > 0$, is an entire function of exponential type ω then for any real sequence $\{x_j\} \subset \mathbb{R}$ such that

$$\inf_{j \neq m} |x_j - x_m| = \alpha > 0$$

the following inequality holds true:

$$\left(\sum_j \left| f(x_j) \right|^p \right)^{1/p} \leqslant c \| f \|_{L_p(\mathbb{R})}, \tag{1.1}$$

E-mail address: pesenson@math.temple.edu.

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were c is independent of f. Another well-known result of Plancherel and Polya (see [7,17,18,21,22]) states that if $x_j = j \in \mathbb{Z}$ is a sequence of integers and $f \in L_p(\mathbb{R})$, $1 \leqslant p \leqslant \infty$, is an entire function of exponential type less than π, then an inequality opposite to (1.1) also holds true

$$\|f\|_{L_p(\mathbb{R})} \leqslant C\left(\sum_{j\in\mathbb{Z}} \left|f(j)\right|^{1/p}\right)^{1/p}, \quad 1 \leqslant p \leqslant \infty,$$

where C is independent on f.

The sequences of real numbers $\{x_j\}$ for which both inequalities hold true

$$\left(\sum_j \left|f(x_j)\right|^p\right)^{1/p} \leqslant c\|f\|_{L_p(\mathbb{R})} \leqslant C\left(\sum_j \left|f(x_j)\right|^p\right)^{1/p}, \quad 1 \leqslant p \leqslant \infty, \tag{1.2}$$

are known as sampling sequences. The corresponding inequalities are known as Plancherel–Polya inequalities [19], or as the frame inequalities [7].

The problem of describing sampling sequences attracted attention of many mathematicians [5,6,8,10,11,13,14]. An optimal result about sampling was obtained by A. Beurling [5] in terms of its famous notion of density for discrete sets of points. It was recently shown in [13] that a sequence $\{x_j\}$ is a sampling sequence for a Paley–Wiener space in $L_2(\mathbb{R})$ if and only if it is a zero set for a certain analytic function.

According to the Paley–Wiener theorem the Fourier transform

$$\hat{f}(\xi) = \mathcal{F}f(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\xi x} f(x)\, dx$$

of any entire function of exponential type $\leqslant \omega$ is a distribution with support in $[-\omega, \omega]$. In the case $p = 2$ the space of entire functions of exponential type ω is denoted by $PW_\omega(\mathbb{R})$ and is called the Paley–Wiener space.

If $p = 2$ then since the support of the Fourier transform $\hat{f}$ is in $[-\omega, \omega]$ the Plancherel theorem shows that the inequalities (1.2) can be written in the following form:

$$C_1 \sum_{j\in\mathbb{Z}} \left| \int_{-\omega}^{+\omega} \hat{f}(\xi) e^{i\xi x_j}\, d\xi \right|^2 \leqslant \int_{-\omega}^{+\omega} \left|\hat{f}(\xi)\right|^2 d\xi \leqslant C_2 \sum_{j\in\mathbb{Z}} \left| \int_{-\omega}^{+\omega} \hat{f}(\xi) e^{i\xi x_j}\, dt \right|^2, \tag{1.3}$$

that means that the functions $\{e^{i\xi x_j}\}$ form a kind of a basis (not necessary orthogonal), which is called a *frame* in the space $L_2([-\omega, \omega])$.

There is a remarkable result of Duffin and Schaeffer [8], that the inequalities (1.3) imply existence of a *dual frame* $\{\theta_j\}$ in the space $L_2([-\omega, \omega])$ such that any function $f \in PW_\omega(\mathbb{R})$ can be reconstructed according to the following formula:

$$f(x) = \sum_{j\in\mathbb{Z}} f(x_j) \mathcal{F}^{-1}\theta_j(x). \tag{1.4}$$

This formula is a generalization of the classical sampling theorem which says, that if $f \in PW_\omega(\mathbb{R})$ then f is completely determined by its values at points $n\Omega$, where $\Omega = \pi/\omega$ and in L_2-sense

$$f(t) = \sum f(n\Omega) \frac{\sin(\omega(t - n\Omega))}{\omega(t - n\Omega)}.$$

In their classical paper [8] Duffin and Schaeffer found sufficient conditions on a sequence of points $\{x_j\}, x_j \in \mathbb{R}$, for which the exponentials $\{e^{ix_j\xi}\}$ form frames in appropriate spaces $L_2([-\omega,\omega])$. In fact it was a far going development of some ideas of Paley and Wiener [14] about irregular sampling.

We consider spaces $E_p^\omega(\mathbb{R}^d)$, $E_2^\omega(\mathbb{R}^d) = PW_\omega(\mathbb{R}^d)$, of entire functions of exponential type $\leqslant \omega$ which belong to $L_p(\mathbb{R}^d)$.

To formulate our main result we will need the following definition.

Definition 1. A set of compactly supported distributions $\{\Phi_j\}$, $j \in \mathbb{N}$, is called sampling sequence in the space $E_p^\omega(\mathbb{R}^d)$, $1 \leqslant p \leqslant \infty$, if there exist two positive constants C_1, C_2 such that for any $f \in E_p^\omega(\mathbb{R}^d)$ the following Plancherel–Polya-type inequality holds true:

$$C_1 \sum_{j\in\mathbb{N}} \bigl|\Phi_j(f)\bigr|^p \leqslant \int_{\mathbb{R}^d} \bigl|f(x)\bigr|^p\,dx \leqslant C_2 \sum_{j\in\mathbb{N}} \bigl|\Phi_j(f)\bigr|^p.$$

In terms of this definition every classical sampling sequence of points $\{x_j\}$ should be identified with a sampling sequence of Dirac measures δ_{x_j}.

The goal of the present article is to develop sufficient conditions (Theorem 3.1) under which a sequence of compactly supported distributions of order $m \geqslant 0$ will be a sampling sequence in the sense of the Definition 1.

Note that we do not assume that the supports of our distributions are disjoint. In Section 4 we develop a reconstruction algorithm for functions f from the Paley–Wiener space $PW_\omega(\mathbb{R}^d)$ by using their samples defined by some sequences of compactly supported distributions of a fixed order $m \in \mathbb{N}$. Our reconstruction formula (4.6) is a generalization of the Duffin–Schaeffer formula (1.4).

All results of the paper are qualitative and a lot should be done in order to approach an optimal result in the spirit of A. Beurling [5]. Such optimal result would be interesting from a theoretical point of view and useful for applications in signal analysis and image processing.

We consider sets of points $\{x_j\}, x_j \in \mathbb{R}^d$, for which there exists $\rho > 0$ such that the following properties hold true. In what follows the notation $B(x,\rho)$ means a ball on $\mathbb{R}^d$ with center $x \in \mathbb{R}^d$ and of radius ρ.

(1) Balls $B(x_j, \rho/4)$ are disjoint.
(2) Balls $B(x_j, \rho/2)$ form a cover of $\mathbb{R}^d$.
(3) Balls $B(x_j, \rho)$ form a cover of $\mathbb{R}^d$ of finite multiplicity N.

Although the first two conditions imply that the cover $B(x_j,\rho)$ has a finite multiplicity, it is important for us to keep multiplicity of this cover bounded by a fixed constant N.

In Lemma 2.1 we prove the existence of such sets. Namely, we show that there exists a certain uniform bound $N_d = N(\mathbb{R}^d)$ on multiplicity of all such covers with sufficiently small ρ's.

Definition 2. We will use notation $X(x_j,\rho,N_d)$ for any set of points $\{x_j\} \in \mathbb{R}^d$ which satisfies the above properties (1)–(3) and we will call such set a (ρ, N_d)-lattice in $\mathbb{R}^d$.

In what follows we will always assume that $0 < \rho < 1$.

For a fixed (ρ, N_d)-lattice $X(x_j,\rho,N_d)$ we consider the following set $\Phi = \{\Phi_j\}$ of distributions Φ_j.

Let $K_j \subset B(x_j, \rho/2)$ be a compact subset and μ_j be a non-negative measure on K_j. We will always assume that the total measure of K_j is finite, i.e.,

$$0 < |K_j| = \int_{K_j} d\mu_j < \infty.$$

We consider the following distribution on $C_0^\infty(B(x_j, \rho))$:

$$\Phi_j(\varphi) = \int_{K_j} \varphi \, d\mu_j, \tag{1.5}$$

where $\varphi \in C_0^\infty(B(x_j, \rho))$. As a compactly supported distribution of order zero it has a unique continuous extension to the space $C^\infty(B(x_j, \rho))$.

We say that a family $\Phi = \{\Phi_j\}$ is uniformly bounded, if there exists a positive constant C_Φ such that

$$|K_j| \leqslant C_\Phi \tag{1.6}$$

for all j.

We will also say that a family $\Phi = \{\Phi_j\}$ is separated from zero if there exists a constant $c_\Phi > 0$ such that

$$|K_j| \geqslant c_\Phi \tag{1.7}$$

for all j.

Some examples of such distributions which are of particular interest to us are the following:

(1) Delta functionals. In this case $K_j = \{x_j\}$, measure $d\mu_j$ is any positive number μ_j and $\Phi_j(f) = \mu_j \delta_{x_j}(f) = \mu_j f(x_j)$.
(2) Finite or infinite sequences of delta functions $\delta_{x_{j,k}}, x_{j,k} \in B(x_j, \rho/2)$, with corresponding weights $\mu_{j,k}$. In this case $K_j = \bigcup_k \{x_{j,k}\}$ and

$$\Phi_j(f) = \sum_k \mu_{j,k} \delta_{x_{j,k}}(f),$$

where we assume the following:

$$0 < |K_j| = \sum_k |\mu_{j,k}| < \infty, \quad K_j = \bigcup_k \{x_{j,k}\}.$$

(3) K_j is a smooth submanifold in $B(x_j, \rho/2)$ of any codimension and $d\mu_j$ is its "surface" measure.
(4) K_j is the closure of $B(x_j, \rho/2)$ and $d\mu_j$ is the restriction of the weighted Lebesgue measure dx on $\mathbb{R}^d$.

Note that reconstruction of band limited functions in $L_2(\mathbb{R}^d)$ from their average values over sets of full measure was first considered in [9] and then later developed in [2,3,15,16,20]. But our approach to the problem seems to be quite different.

Our ultimate goal is to consider families of distributions which are more general than the family $\{\Phi_j\}$. Namely, we will be interested in families of compactly supported distributions of order $m \geqslant 0$ which have the form $\Psi_j = A\Phi_j$, where the family Φ_j is described in (1.5)–(1.7) and A is an invertible linear differential operator of order m.

For such family of distributions we prove our main Theorem 3.1 bellow which states that for any invertible differential operator A, any constants $C_\Phi > c_\Phi > 0$, there exist positive constants C, C_1, c_1, such that for every $\omega > 0$, every (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $0 < \rho < (C\omega)^{-1}$, every family of distributions $\{\Phi_j\}$ of the form (1.5) with properties (1.6), (1.7) and every $f \in E_p^\omega(\mathbb{R}^d)$ the following inequalities hold true:

$$c_1\left(\sum_i \left|\Psi_j(f)\right|^p\right)^{1/p} \leqslant \rho^{-d/p}\|f\|_p \leqslant C_1\left(\sum \left|\Psi_j(f)\right|^p\right)^{1/p}, \quad \Psi_j = A\Phi_j,$$

for all $f \in E_p^\omega(\mathbb{R}^d)$.

2. Poincaré type inequalities in $L_p(\mathbb{R}^d)$

In what follows the notation $W_p^k(\mathbb{R}^d)$, $k \in \mathbb{N}$, $W_p^0(\mathbb{R}^d) = L_p(\mathbb{R}^d)$, $1 \leqslant p \leqslant \infty$, is used for Sobolev spaces of all functions in $L_p(\mathbb{R}^d)$ whose distributional mixed derivatives up to the order k belong to $L_p(\mathbb{R}^d)$ and the norm is defined as

$$\|f\|_{W_p^k(\mathbb{R}^d)} = \|f\|_p + \sum_{|\alpha|\leqslant k} \left\|\partial^{|\alpha|} f\right\|_p,$$

were $\|f\|_p$ means the $L_p(\mathbb{R}^d)$-norm of the function f. This norm is equivalent to each of the following norms (see [1,12]):

$$\|f\|_p + \sum_{|\alpha|=k} \left\|\partial^{|\alpha|} f\right\|_p$$

or

$$\|f\|_p + \left\|\Delta^{k/2} f\right\|_p,$$

where Δ is the Laplace operator in $L_p(\mathbb{R}^d)$.

In the case $p = 2$ we use the notation $W_2^k(\mathbb{R}^d) = H^k(\mathbb{R}^d)$. We will need the following lemma.

Lemma 2.1. *For any $\rho > 0$ there exists a set of points $\{x_j\}$ in $\mathbb{R}^d$ such that*

(1) *balls $B(x_j, \rho/4)$ are disjoint,*
(2) *balls $B(x_j, \rho/2)$ form a cover of $\mathbb{R}^d$,*
(3) *multiplicity of the cover by balls $B(x_j, \rho)$ is not greater $N_d = 12^d$.*

Proof. The last statement means that every point of $\mathbb{R}^d$ belong to no more than $N_d = 12^d$ balls from the family $B(x_j, \rho)$.

Let us choose a family of disjoint balls $B(x_j, \rho/4)$ such that there is no ball $B(x, \rho/4)$, $x \in \mathbb{R}^d$, which has empty intersections with all balls from our family. Then the family $B(x_j, \rho/2)$ is a cover of $\mathbb{R}^d$. Every ball from the family $\{B(x_j, \rho)\}$, that has non-empty intersection with a particular ball $B(x_k, \rho)$ is contained in the ball $B(x_k, 3\rho)$. Since any two balls from the family $\{B(x_j, \rho/4)\}$ are disjoint, it gives the following estimate for the index of multiplicity N of the cover $\{B(x_j, \rho)\}$:

$$N \leqslant \frac{\operatorname{Vol} B(x, 3\rho)}{\operatorname{Vol} B(x, \rho/4)} = 12^d. \qquad \square$$

First we prove a local Poincaré-type inequality. In the following lemma we consider a fixed ball $B(x_0, \rho)$ and a fixed compact set $K \subset B(x_0, \rho)$ with a measure μ on it. The corresponding functional Φ is defined by the formula (1.5).

Lemma 2.2. *For any $k > d/p$ there exists a constant $C = C(d, k) > 0$, such that for any ball $B(x_0, \rho)$, $x_0 \in \mathbb{R}^d$, any distribution Φ of type* (1.5) *the following inequality holds true*:

$$\left\| f - |K|^{-1}\Phi(f) \right\|_{L_p(B(x_0,\rho/2))} \leqslant C(d,k) \sum_{1\leqslant|\alpha|\leqslant k} \rho^{|\alpha|} \left\| \partial^{|\alpha|} f \right\|_{L_p(B(x_0,\rho))}, \tag{2.1}$$

where $f \in W_p^k(\mathbb{R}^d)$, $k > d/p$, $1 \leqslant p \leqslant \infty$, and $\partial^m f$ is a partial derivative of order m in a rectangular coordinate system in $\mathbb{R}^d$.

Proof. For any $f \in C^\infty(\mathbb{R}^d)$, every $x, y \in B(x_0, \rho/2)$ we have the following:

$$f(x) = f(y) + \sum_{1\leqslant|\alpha|\leqslant k-1} \frac{1}{\alpha!} \partial^{|\alpha|} f(y)(x-y)^\alpha + \sum_{|\alpha|=k} \frac{1}{\alpha!} \int_0^\eta t^{k-1} \partial^{|\alpha|} f(y + t\vartheta)\vartheta^\alpha \, dt,$$

where $x = (x_1, \ldots, x_d)$, $y = (y_1, \ldots, y_d)$, $\alpha = (\alpha_1, \ldots, \alpha_d)$, $(x-y)^\alpha = (x_1 - y_1)^{\alpha_1} \cdots (x_d - y_d)^{\alpha_d}$, $\eta = \|x - y\|$, $\vartheta = (x-y)/\eta$.

We integrate each term over compact $K \subset B(x_0, \rho)$ against $d\mu(y)$, where $d\mu$ is the measure on K. It gives

$$\begin{aligned} &f(x) - |K|^{-1}\Phi(f) \\ &\quad = |K|^{-1} \int_K \left(\sum_{1\leqslant|\alpha|\leqslant k-1} \frac{1}{\alpha!} \partial^{|\alpha|} f(y)(x-y)^\alpha \right) d\mu(y) \\ &\qquad + |K|^{-1} \int_K \left(\sum_{|\alpha|=k} \frac{1}{(k-1)!} \int_0^\eta t^{k-1} \partial^{|\alpha|} f(y + t\vartheta)\vartheta^\alpha \, dt \right) d\mu(y), \end{aligned}$$

and then

$$\begin{aligned} &\left| f(x) - |K|^{-1}\Phi(f) \right|^p \\ &\quad \leqslant C(k,d)|K|^{-1} \sum_{1\leqslant|\alpha|\leqslant k-1} \left(\int_K \left| \partial^{|\alpha|} f(y)(x-y)^\alpha \right| d\mu(y) \right)^p \\ &\qquad + C(k,d)|K|^{-1} \sum_{|\alpha|=k} \left(\int_K \left| \int_0^\eta t^{k-1} \partial^{|\alpha|} f(y + t\vartheta)\vartheta^\alpha \, dt \right| d\mu(y) \right)^p. \end{aligned}$$

Thus, we obtain

$$\begin{aligned} &\left\| f - |K|^{-1}\Phi(f) \right\|_{L_p(B(x_0,\rho/2))} \\ &\quad \leqslant C(k,d)|K|^{-1} \sum_{1\leqslant|\alpha|\leqslant k-1} \left(\int_{B(x_0,\rho/2)} \left(\int_K \left| \partial^{|\alpha|} f(y)(x-y)^\alpha \right| d\mu(y) \right)^p dx \right)^{1/p} \end{aligned}$$

$$+ C(k,d)|K|^{-1} \sum_{|\alpha|=k} \left(\int\limits_{B(x_0,\rho/2)} \left(\int\limits_K \left| \int\limits_0^\eta t^{k-1} \partial^{|\alpha|} f(y+t\vartheta)\vartheta^\alpha \, dt \right| d\mu(y) \right)^p dx \right)^{1/p}. \tag{2.2}$$

By the Minkowski inequality the first term on the right in (2.2) is not greater than

$$C(k,d)|K|^{-1} \sum_{1\leqslant|\alpha|\leqslant k-1} \int\limits_K \left(\int\limits_{B(x_0,\rho/2)} \left| \partial^{|\alpha|} f(y)(x-y)^\alpha \right|^p dx \right)^{1/p} d\mu(y). \tag{2.3}$$

Now we are going to use the known inequality [1]

$$\left|\psi(y)\right| \leqslant \sum_{0\leqslant n\leqslant m} C(d,m,n)\rho^{n-d/p} \|\psi\|_{W_p^n(B(x_0,\rho))}, \quad m > d/p,$$

where $y \in B(x_0,\rho/2)$, $\psi \in C^\infty(B(x_0,\rho))$. This inequality implies the inequality

$$\left|\partial^{|\alpha|} f(y)\right| \leqslant C(d,m) \sum_{0\leqslant|\sigma|\leqslant m} \rho^{|\sigma+\alpha|-d/p} \left\| \partial^{|\sigma+\alpha|} f \right\|_{L_p(B(x_0,\rho))}, \tag{2.4}$$

where $y \in B(x_0,\rho/2)$, $\sigma = (\sigma_1,\sigma_2,\ldots,\sigma_d)$, $m > d/p$.

A combination of (2.3) and (2.4) shows that the first term in (2.2) is not greater than

$$C(k,d) \sum_{1\leqslant|\gamma|\leqslant m+k-1} \rho^{|\gamma|} \left\| \partial^{|\gamma|} f \right\|_{L_p(B(x_0,\rho))}.$$

By the same Minkowski inequality the second term on the right in (2.2) is not greater than

$$C(k,d)|K|^{-1} \sum_{|\alpha|=k} \int\limits_K \left(\int\limits_{B(x_0,\rho/2)} \left| \int\limits_0^\eta t^{k-1} \partial^{|\alpha|} f(y+t\vartheta)\vartheta^\alpha \, dt \right|^p dx \right)^{1/p} d\mu(y).$$

By using the Hölder inequality and the assumption $k > d/p$ we estimate the inner integral

$$\left| \int\limits_0^\eta t^{k-1} \partial^{|\alpha|} f(y+t\vartheta)\vartheta^\alpha \, dt \right|^p \leqslant C\tau^{pk-d} \int\limits_0^\eta t^{d-1} \left| \partial^{|\alpha|} f(y+t\vartheta) \right|^p dt.$$

So, we obtain that the second term on the right side of (2.2) is not greater than

$$C(k,d)|K|^{-1} \left(\int\limits_K \left(\int\limits_{B(x_0,\rho/2)} \eta^{pk-d} \int\limits_0^\eta t^{d-1} \left| \partial^{|\alpha|} f(y+t\vartheta) \right|^p dt \right) dx \right)^{1/p}.$$

We integrate over the ball $B(x_0,\rho/2)$ using the spherical coordinate system (η,ϑ) and since $\eta \leqslant \rho$ we obtain that the above integral is not larger than the term

$$C(k,d)\rho^{pk} \sum_{|\alpha|=k} \left\| \partial^{|\alpha|} f \right\|^p_{L_p(B(x_0,\rho))}.$$

The above estimates give the desired inequality (2.1). □

Lemma 2.3. *For any $k > d/p$, $C_\Phi > 0$, there exists a constant $C = C(C_\phi, d, k)$ such that for any lattice $X(x_j, \rho, N_d)$, $0 < \rho < 1$, any family of distributions $\Phi = \{\Phi_j\}$ defined in* (1.5) *with the property* (1.6) *and any $f \in W_p^k(\mathbb{R}^d)$ the following inequality holds true*:

$$\left(\sum_j \left|\Phi_j(f)\right|^p\right)^{1/p} \leqslant C\rho^{-d/p}\|f\|_{W_p^k(\mathbb{R}^d)}, \quad k > d/p,\ 1 \leqslant p \leqslant \infty. \tag{2.5}$$

Proof. Using properties of distributions Φ_j and the inequality

$$\left|\psi(y)\right| \leqslant C_0(d,k) \sum_{0\leqslant m\leqslant k} \rho^{m-d/p}\|\psi\|_{W_p^m(B(x_j,\rho))}, \quad k > d/p,$$

where $y \in B(x_j, \rho/2)$, $\psi \in C^\infty(B(x_j, \rho))$, we obtain

$$\left|\Phi_j(f)\right| \leqslant C_\Phi \sup_{x\in B(x_j,\rho/2)} \left|f(x)\right| \leqslant C_0 C_\Phi \sum_{0\leqslant m\leqslant k} \rho^{m-d/p}\|f\|_{W_p^m(B(x_j,\rho))},$$

where $f \in C^\infty(\mathbb{R}^d) \cap L_p(\mathbb{R}^d)$, $k > d/p$, $C_0 = C_0(d,k)$, and then

$$\left(\sum_j \left|\Phi_j(f)\right|^p\right)^{1/p} \leqslant C_0 C_\Phi \left(\sum_j \sum_{0\leqslant m\leqslant k} \rho^{m-d/p}\|f\|^p_{W_p^m(B(x_j,\rho))}\right)^{1/p}.$$

Since the multiplicity of the cover by balls $B(x_j, \rho)$ is not greater N_d we obtain

$$\begin{aligned}\left(\sum_j \left|\Phi_j(f)\right|^p\right)^{1/p} &\leqslant C\rho^{-d/p}\left(\sum_{0\leqslant m\leqslant k} \sum_j \|f\|^p_{W_p^m(B(x_j,\rho))}\right)^{1/p} \\ &= C\rho^{-d/p}\|f\|_{W_p^k(\mathbb{R}^d)},\end{aligned}$$

where $C = C(C_\Phi, d, k)$. □

Our global Poincaré-type inequality is the following.

Theorem 2.4. *For any $k > d/p$, $C_\Phi > c_\Phi > 0$, there exist constants $C = C(d, C_\Phi, c_\Phi, k) > 0$ and $\rho_0 = \rho_0(d, C_\Phi, c_\Phi, k) > 0$, such that for any (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $\rho < \rho_0$, any family of distributions $\Phi = \{\Phi_j\}$ of the form* (1.5) *which satisfy* (1.6) *and* (1.7) *with the constants C_Φ and c_Φ the following inequality holds true*:

$$\|f\|_p \leqslant C\left\{\rho^{d/p}\left(\sum_j |K_j|^{-1}\left|\Phi_j(f)\right|^p\right)^{1/p} + \rho^k\left\|\Delta^{k/2} f\right\|_p\right\}, \quad 1 \leqslant p \leqslant \infty, \tag{2.6}$$

for all $f \in W_p^k(\mathbb{R}^d)$.

Proof. Since the balls $B(x_j, \rho)$ form a cover of $\mathbb{R}^d$ of finite multiplicity the inequality (2.1) implies the inequality

$$\sum_{j=1}\left\|f - |K_j|^{-1}\Phi_j(f)\right\|_{L_p(B(x_j,\rho/2))} \leqslant C(d,k)C_\Phi c_\Phi^{-1} \sum_{1\leqslant\nu\leqslant k} \rho^\nu \|f\|_{W_p^\nu(\mathbb{R}^d)}. \tag{2.7}$$

Next, we have

$$\|f\|_p^p \leqslant \sum_j \|f\|^p_{L_p(B(x_j,\rho/2))}$$
$$\leqslant 2\sum_j \big\|f - |K_j|^{-1}\Phi_j(f)\big\|^p_{L_p((B(x_j,\rho/2))} + 2\,\mathrm{Vol}\big(B(x_j,\rho/2)\big)\sum_j |K_j|^{-1}\big|\Phi_j(f)\big|^p,$$

and then

$$\|f\|_p \leqslant C(d,k)C_\Phi c_\Phi^{-1}\left\{\rho^{d/p}\left(\sum_j |K_j|^{-1}\big|\Phi_j(f)\big|^p\right)^{1/p} + \sum_{1\leqslant\nu\leqslant k}\rho^\nu\|f\|_{W_p^\nu(\mathbb{R}^d)}\right\}.$$

Since

$$\|f\|_{W_p^\nu(\mathbb{R}^d)} \leqslant c_\nu\big(\|f\|_p + \big\|\Delta^{\nu/2}f\big\|_p\big)$$

we obtain

$$\|f\|_p \leqslant C\left\{\rho^{d/p}\left(\sum_{j=1}^\infty |K_j|^{-1}\big|\Phi_j(f)\big|^p\right)^{1/p} + \sum_{\nu=1}^k \rho^\nu\big(\|f\|_p + \big\|\Delta^{\nu/2}f\big\|_p\big)\right\}, \tag{2.8}$$

where $k > d/p$, $C = C(C_\Phi, c_\Phi, d, k)$.

Since according to Lemma 2.1 for any $\rho > 0$ there are (ρ, N_d)-lattices we can assume that ρ in the last inequality is so small that

$$\rho \times k \times C(C_\Phi, c_\Phi, d, k) = \frac{1}{2}.$$

For any (ρ, N_d)-lattice with such ρ the inequality (2.8) implies

$$\|f\|_p \leqslant C'(C_\Phi, c_\Phi, d, k)\left\{\rho^{d/p}\left(\sum_{j=1}^\infty |K_j|^{-1}\big|\Phi_j(f)\big|^p\right)^{1/p} + \sum_{\nu=1}^k \rho^\nu\big\|\Delta^{\nu/2}f\big\|_p\right\}. \tag{2.9}$$

According to the interpolation inequality for any $a > 0$, $\rho > 0$, $0 \leqslant \nu \leqslant k$ we have

$$\rho^\nu\big\|\Delta^{\nu/2}f\big\|_p \leqslant a^{k-\nu}\rho^k\big\|\Delta^{k/2}f\big\|_p + c_k a^{-\nu}\|f\|_p, \quad 1 \leqslant \nu \leqslant k.$$

Combination of this inequality with the inequality (2.9) shows that we can choose a large a such that the inequality (2.6) will be obtained. □

3. Plancherel–Polya-type inequalities for functions from $E_p^\omega(\mathbb{R}^d)$

By using our family $\{\Phi_j\}$ of distributions of order 0 which is defined in (1.5)–(1.7) and a differential operator A of order $m \in \mathbb{N}$ which has bounded inverse in $L_p(\mathbb{R}^d)$ we introduce a family of distributions of order m. Namely, we will use notation Ψ_j for a compactly supported distribution of order $m \in \mathbb{N}$, which has the form

$$\Psi_j = A\Phi_j. \tag{3.1}$$

This definition simply means that

$$\Psi_j(f) = \int_{K_j} Af\,d\mu_j,$$

where the compact set K_j and the measure μ_j on it are from the definition (1.5) of the distribution Φ_j.

Our next goal is to prove a Plancherel–Polya inequality for a family of distributions $\{\Psi_j\}$.

Theorem 3.1. *For any given invertible differential operator A, any given $C_\Phi > c_\Phi > 0$, there exist positive constants C, C_1, c_1, such that for every $\omega > 0$, every (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $0 < \rho < (C\omega)^{-1}$, every family of distributions $\{\Psi_j\}$ of the form* (3.1) *where a family $\{\Phi_j\}$ is defined in* (1.5)–(1.7) *with the given $C_\Phi \geqslant c_\Phi$ the following inequalities hold true*:

$$c_1\rho^{d/p}\left(\sum\left|\Psi_j(f)\right|^p\right)^{1/p} \leqslant \|f\|_p \leqslant C_1\rho^{d/p}\left(\sum\left|\Psi_j(f)\right|^p\right)^{1/p}, \quad 1 \leqslant p \leqslant \infty,$$

for every $f \in E_p^\omega(\mathbb{R}^d)$.

Proof. First, we are going to prove the above inequality for the family of measures $\{\Phi_j\}$. By Lemma 2.4 there exists C such that for any (ρ, N_d)-lattice with sufficiently small ρ we have the inequality for $k_d = [d/2] + 1$,

$$\|f\|_p \leqslant C\left\{\rho^{d/p}\left(\sum_{j=1}^{\infty}\left|\Phi_j(f)\right|^p\right)^{1/p} + \rho^{k_d}\left\|\Delta^{k_d/2}f\right\|_p\right\}.$$

If for the fixed ω and k_d we choose ρ such that

$$Cd^{k_d}\omega^{k_d}\rho^{k_d} \leqslant \frac{1}{2},$$

then since every function from $E_p^\omega(\mathbb{R}^d)$ satisfies the Bernstein inequality

$$\left\|\Delta^{k_d/2}f\right\|_p \leqslant (d\omega)^{k_d}\|f\|_p$$

for any $f \in E_p^\omega(\mathbb{R}^d)$ and for any $k = 0, 1, 2, \ldots$, we will have the inequality

$$\|f\|_p \leqslant C_1\rho^{d/2}\left(\sum_{j=1}^{\infty}\left|\Phi_j(f)\right|^p\right)^{1/p}.$$

On the other hand, by Lemma 2.3 for k_d,

$$\rho^{d/p}\left(\sum_{j=1}^{\infty}\left|\Phi_j(f)\right|^p\right)^{1/p} \leqslant C\|f\|_{W_d^k(\mathbb{R}^d)} \leqslant C(d)\max\left(1, \omega^{k_d}\right)\|f\|_p = c_1\|f\|_p.$$

Now, the theorem follows from the Bernstein inequality for functions in $E_p^\omega(\mathbb{R}^d)$ and invertibility of the operator A. □

For a set of distributions $\Psi = \{\Psi_j\}$ and $\omega > 0$ the notation $l_p^\omega(\Psi)$ will be used for a linear subspace of all sequences $\{v_j\}$ in l_p for which there exists a function f in $PW_\omega(\mathbb{R}^d)$ such that

$$\left\{\Psi_j(f)\right\} = v_j, \quad j \in \mathbb{N}.$$

In general $l_p^\omega(\Psi) \neq l_p$.

Definition 3. A linear reconstruction method R from a set of distributions Ψ is a linear operator

$$R : l_p^\omega(\Psi) \to E_p^\omega\left(\mathbb{R}^d\right)$$

such that

$$R : \left\{\Psi_j(f)\right\} \to f.$$

The reconstruction method is said to be stable, if it is continuous in topologies induced respectively by l_p and $L_p(\mathbb{R}^d)$.

The following statement is a direct consequence of Theorem 3.1.

Theorem 3.2. *For any given $C_\Phi > c_\Phi > 0$ and any invertible differential operator A there exist positive constants C, C_1, c_1, such that for every $\omega > 0$, every (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $0 < \rho < (C\omega)^{-1}$, every family of distributions $\{\Psi_j = A\Phi_j\}$ where Φ_j are of the form* (1.5) *with properties* (1.6), (1.7) *the following statements hold true*:

(1) *every function f in $E_p^\omega(\mathbb{R}^d)$, $1 \leqslant p \leqslant \infty$, is uniquely determined by the set of samples $\{\Psi_j(f)\}$;*
(2) *a reconstruction method R from a such set of samples $\{\Psi_j(f)\}$,*

$$R : \left\{\Psi_j(f)\right\} \to f$$

is stable.

4. Reconstruction in terms of frames

In this section we consider the case $p = 2$. We discuss a method of reconstruction of a function f in the space $PW_\omega(\mathbb{R}^d)$ from a set of samples $\{\Psi_j(f)\}$, where distributions Ψ_j are the distributions (3.1) and where Φ_j is the family of measures described in (1.5)–(1.7). In this section we assume that the invertible differential operator A in the formula (3.1) is fixed.

We consider the Fourier transform

$$\widehat{\Psi_j}(\xi) = \Psi_j\left(e^{i\langle x,\xi\rangle}\right) \tag{4.1}$$

of the distribution Ψ_j. From Theorem 3.1 we obtain

$$c\|\hat{f}\|_2 \leqslant \left(\sum_j \left|\langle\widehat{\Psi_j}, \hat{f}\rangle\right|^2\right)^{1/2} \leqslant C\|\hat{f}\|_2, \tag{4.2}$$

where $\langle .,. \rangle$ is the scalar product in the space $L_2(B(0,\omega))$. The following statement is just another interpretation of the inequalities (4.2).

Theorem 4.1. *For a given invertible differential operator A, any given constants $C_\Phi > c_\Phi > 0$, there exists a positive constant C such that for every $\omega > 0$, every (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $0 < \rho < (C\omega)^{-1}$, every family of distributions $\Psi_j = A\Phi_j$ where Φ_j are defined in* (1.5)–(1.7)*, the restrictions of the holomorphic functions $\{\widehat{\Psi_j}\}$ to the ball $(B(0,\omega))$ form a frame in the space $L_2(B(0,\omega))$.*

In the important particular case of point-wise sampling on a lattice $X(x_j, \rho, N_d)$ the formula (4.1) takes the form

$$\widehat{\Psi_j}(\xi) = e^{i\langle x_j, \xi\rangle}. \tag{4.3}$$

In this situation the last theorem means that the Fourier transforms of $\delta_{x_j}, x_j \in Z$, form a frame in the space $L_2(B(0,\omega))$.

According to the general theory [4,8] there exists a dual frame $\widehat{\Theta_j}$ in $L_2(B(0,\omega))$ and a reconstruction formula can be written in terms of the dual frame as

$$\hat{f} = \sum_j \langle \widehat{\Theta_j}, \hat{f}\rangle \widehat{\Psi_j} = \sum_j \langle \widehat{\Psi_j}, \hat{f}\rangle \widehat{\Theta_j}, \tag{4.4}$$

where inner product is taken in the space $L_2(B(0,\omega))$.

To describe the dual frame one has to consider the so called frame operator

$$F(\hat{f}) = \sum_j \langle \widehat{\Psi_j}, \hat{f}\rangle \widehat{\Psi_j}.$$

The operator F is invertible and the dual frame $\{\widehat{\Theta_j}\}$ in the space $L_2(B(0,\omega))$ is introduced by using the following formula:

$$\widehat{\Theta_j} = F^{-1}\widehat{\Psi_j}. \tag{4.5}$$

Theorem 4.2. *For a given invertible differential operator A, any given constants $C_\Phi > c_\Phi > 0$, there exists a positive constant C such that for every $\omega > 0$, every (ρ, N_d)-lattice $X(x_j, \rho, N_d)$ with $0 < \rho < (C\omega)^{-1}$, every family of distributions $\Psi_j = A\Phi_j$ where Φ_j are defined in* (1.5)–(1.7)*, every Paley–Wiener function $f \in PW_\omega(\mathbb{R}^d)$ can be reconstructed from the set of samples $\{\Psi_j(f)\}$ by using the formula*

$$f = \sum_j \Psi_j(f)\Theta_j, \tag{4.6}$$

where $\{\Theta_j\}$ is a frame in $PW_\omega(\mathbb{R}^d)$ and the Fourier transforms $\widehat{\Theta_j}$ of the functions Θ_j are defined in (4.5)*.*

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